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12 March 1984
SUPERSEDING
MIL-S-43880B
15 March 1979

MILITARY SPECIFICATION

SLEEPING BAG, EXTREME COLD WEATHER (DOWN AND POLYESTER BATTING)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers the requirements for one type and size of sleeping bag for extreme cold conditions.

2. APPLICABLE DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

SPECIFICATIONS

FEDERAL

P-S-1792	-	Soap, Laundry (Neutral and Built)
V-F-106	-	Fasteners, Slide, Interlocking
V-L-61	-	Laces, Nylon
AAA-S-108	-	Scale, Dial and Beam Indicating, Bench (5 to 200 Pound Capacity)
DDD-L-20	-	Label: For Clothing, Equipage, and Tentage (General Use)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research and Development Center, Natick, MA 01760, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8465

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DDD-T-86 - Tape, Textile; Cotton, General Purpose (Unbleached, Bleached, or Dyed)
PPP-B-636 - Boxes, Shipping, Fiberboard

MILITARY

MIL-C-332 - Cloth, Balloon, Cotton
MIL-W-530 - Webbing, Textile, Cotton, General Purpose, Natural or in Colors
MIL-T-3530 - Thread and Twine; Mildew Resistant or Water Repellent Treated
MIL-F-10884 - Fasteners, Snap
MIL-B-41826 - Batting, Synthetic Fibers: Polyester, (Quilted and Unquilted)
MIL-F-43097 - Feathers and Down, Waterfowl; Tan-O-Quil-QM Treated
MIL-T-43548 - Thread, Polyester, Cotton-Covered and Rayon Covered
MIL-T-43566 - Tape, Textile, Cotton, General Purpose, Natural or in Colors
MIL-C-43876 - Cloth, Wind Resistant Oxford, Nylon and Cotton, Quarpel Treated
MIL-H-43879 - Hood, Sleeping Bag
MIL-W-44049 - Webbing, Textile and Tape, Textile, Polypropylene, General Purpose, Natural or In Colors

STANDARDS

FEDERAL

FED-STD-751 - Stitches, Seams, and Stitchings
FED-STD-191 - Textile Test Methods

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads

DRAWINGS

U.S. ARMY NATICK RESEARCH AND DEVELOPMENT CENTER

2-2-325 - Sleeping Bag, Down and Polyester Batting; Assembly and Sections
2-2-327 - Sleeping Bag, Down and Polyester Batting; Inner and Outer Panels

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2-2-328	-	Sleeping Bag, Down and Polyester Batting; Channel Stitching Arrangements on Panels
2-2-329	-	Sleeping Bag, Down and Polyester Batting; Slide Fastener Assembly and Weatherstrip
2-2-330	-	Sleeping Bag, Down and Polyester Batting; Front Panel Assembly
2-2-362	-	Sleeping Bag, Down and Polyester Batting; Flap Assembly

(Copies of documents required by manufacturers in connnection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

COLOR ASSOCIATION OF THE UNITED STATES, INC.

Department of Defense Standards Shades for Sewing Threads 1968

(Application for copies should be addressed to the Color Association of the United States, Inc., 24 East 38th Street, New York, NY 10016.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

* 2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

3. REQUIREMENTS

3.1 Guide sample. Samples, when furnished, are solely for guidance and information to the contractor (see 6.3). Variations from this document may appear in the sample, in which case this document shall govern.

3.2 First article. When specified, a sample shall be subjected to first article inspection (see 4.3, 6.2 and 6.4).

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3.3 Materials and components. Materials (see 6.7) and components shall be as specified herein and on the drawings. For materials or components for which it is stated "or equal" if the contractor proposes to use an item considered to be equal to the material or component specified. Prior to its use, the contractor shall furnish a sample of the material or component with supporting data to the contracting officer for subsequent evaluation by the responsible agency.

3.3.1 Cloth, balloon, cotton. The cotton balloon cloth shall be Olive Green 107, durable water repellent treated, preshrunk, conforming to type I, class 2 of MIL-C-332 except the shrinkage requirement may be up tn 3.5 percent in the warp and 2.5 percent in the filling.

* 3.3.2 Cloth, non-woven. The material for the diaphragms for the outer panels shall be either non-woven spunbonded nylon cloth conforming to Monsanto CEREX 1.5 to 1.8 ounces per square yard, or equal.

3.3.3 Cloth, oxford, nylon and cotton. The nylon and cotton oxford cloth for the outer panels and drawstring casing shall be quarpel treated, color Olive Green 107 conforming to MIL-C-43876.

3.3.4 Tape, cotton, 1-1/4 inches wide. The 1-1/4 inches wide tape for binding the face opening shall conform to type IV, class 3 of DDD-T-86. The color shall be Olive Drab 7 to match the standard sample (see 6.3). The dyed tape shall show fastness to crocking, laundering and accelerated weathering.

3.3.5 Lace, nylon. The nylon lace for the face closure drawstring shall conform to type I, class 2, Olive Green 107 of V-L-61 except in lieu of applying tips to the ends, the ends shall be fused (see 3.6.4).

3.3.6 Tape, cotton, 3/4 and 2-1/2 inches wide. The 3/4 inch wide tape for the snap fastener retainers, the 3/4 inch wide tape for the footstraps and front opening reinforcement, and the 2-1/2 inch wide tape for the weatherstrip shall conform to type I, class 8, Olive Drab 7 of MIL-T-43566.

3.3.7 Webbing, cotton or polyester 1-1/4 inch wide. The webbing for the slide fastener guard strap may be either cotton webbing Olive Drab 7 conforming to type IIa, class 8, 1-1/4 inch wide of MIL-W-530 or polypropylene webbing conforming to type IIa, class 4, 1-1/4 inch wide of MIL-W-44049.

3.3.8 Thread, polyester, cotton-covered and rayon-covered. The thread for seaming and stitching shall conform to ticket No. 30, 3 ply; ticket No. 50, 2 ply; and ticket No. 70, 2 ply of MIL-T-43548.

3.3.8.1 Color and treatment. The thread for all stitching shall be dyed Olive Drab S-1, C.A. 66022. All thread shall be water repellent treated in accordance with type II, class 3 of MIL-T-3530. The dyed thread shall show fastness to laundering equal to or better than the standard sample.

3.3.9 Batting, polyester. The batting used in the sleeping bag shall be as specified below.

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3.3.9.1 Weatherstrip and flap. The batting for the weatherstrip and flap shall conform to type V of MIL-B-41826 (see 3.4.1).

3.3.9.2 Inner panels. The batting for the inner panels shall conform to type VI of MIL-B-41826 (see 3.4.1).

3.3.10 Down. The filling for the outer panels of the bag shall consist of 100 percent chemically modified waterfowl down conforming to type II, class 1 of MIL-F-43097.

3.3.11 Fasteners, snap. The snap fasteners shall conform to style 2, finish 2 of MIL-F-10884. The male component shall be the stud with size 1 eyelet and the female component shall be the socket with a 24 line size, size 1 button.

* 3.3.12 Fastener, slide, interlocking. The slide fastener shall be type III, style 1, size MHS of V-F-106 except that the chain shall be nylon or polyester cotinuous monofilament in a coil or ladder configuration, or shall be an extruded nylon chain of separate elements connected with strands of polyester yarns, conforming to the requirements stated in 3.3.12.1.1 or 3.3.12.1.2, as applicable. All performance requirements governing the chain are not applicable except the crosswise breaking strength requirement which shall be 175 pounds minimum. The crosswise breaking srength shall be performed as specified in V-F-106 except the fastener shall be preconditioned.

* 3.3.12.1 Fastener chains.

* 3.3.12.1.1 Continuous filament chain. The diameter of the chain filament shall be 0.333 to 0.050 inch. The width of the chain when closed shall be 0.320 to 0.340 inch. The chain shall be securely sewn to the tapes.

* 3.2.12.1.2 Extruded chain. The extruded chain shall consist of a series of nylon elements connected by polyester yarns. The chain when closed shall meet the size requirements specified in V-F-106. The chain shall be securely sewn to the tapes.

3.3.12.1.3 Color. The color of the chains shall be black or olive drab.

3.3.12.2 Fastener stops. The stops shall be plastic (nylon), brass, aluminum, or other non-corroding metal color black or olive drab. The closed bottom stop shall enclose the sides and ends of the fastener chain. All performance requirements governing the stops are not applicable.

3.3.12.3 Fastener length. The fastener length shall be such as to provide a smooth and secure closure and meet the requirements specified on Drawing 2-2-329.

3.3.12.4 Fastener slider and stirrup pulls. The fastener shall have a reversible double pull slider with wire stirrup pulls and thongs attached. The slider shall properly fit the fastener chain. The stirrup pulls shall be fabricated from brass or corrosion resistant steel wire, 0.058 inches minimum diameter. The color of the slider and stirrup pulls shall be black or olive drab.

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3.3.12.5 Slide fastener tape. The slide fastener tape shall be 1 ± 1/16 inch wide, dyed to match Olive Drab S, C.A. 66519, and water repellent treated. The tape shall show fastness to laundering equal to or better than the standard sample (see 6.3). When no standard sample is available, the tape shall show good fastness to laundering.

3.3.13 Hood. The sleeping bag shall be furnished with a hood conforming to MIL-H-43879.

3.3.14 Labels and markings. The labels for the instructions for use and maintenance, laundry instructions, and identification shall be a sewn-on or heat-sealed-on label. The direct marking specified in 3.3.14.1 and the labels for the instructions for use and maintenance, laundry instructinns and identification, shall conform to DDD-L-20 in the types and classes specified herein. If the heat sealed label is used, the modifications specified in 3.3.14.5 shall be made applicable. The labels for the instructions for use and maintenance, laundry instructions and identification shall be printed on one continuous length of cloth and shall be in the order shown on Drawing 2-2-328. The width of the label shall be not more than 5 inches and the length shall be governed by the contents. The sewn on or heat sealed label shall approximate Olive Green 107 as used in the base material for the inner panels specified in 3.3.1 and shall show colorfastness to laundering.

3.3.14.1 Marking. The letters "US" shall be applied to the sleeping bag in the location shown on Drawing 2-2-325 and shall conform to type III or IV, class 9 in solid black letters two inches high minimum (see 6.5). In addition, the following markings conforming to type III or IV, class 9 shall be applied on both sides of the sleeping bag in the location shown on Drawing 2-2-325 in characters 1/2 inch high minimum:

Extreme Cold Weather

The words "Extreme Cold Weather" shall not be abbreviated. The fastness of the class 9 markings shall be as specified for class 5 labels.

3.3.14.2 Instructions for use and maintenance. The instructions for use and maintenance shall conform to type VI, class 3, (except as modified in 3.3.14.5 if a heat-sealed label is used) in characters 3/32 to 1/8 inch high. All words shown in capital letters shall be 3/16 inch high minimum. Instructions for use and maintenance shall be as follows:

INSTRUCTIONS FOR USE

1. Keep the bag dry:

a. The outside fabric of the sleeping bag is water repellent and will protect against moisture. SELECT the driest ground and if possible keep the bag out of the rain.

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b. Breathe through face opening to prevent moisture from wetting the bag. If face is cold, reduce face opening by pulling drawstrings at head end. DO NOT TIE DRAWSTRINGS.

c. DO NOT wear damp clothing and AVOID sweating in bag. If too warm, open slide fastener for ventilation.

d. Open the bag completely and air thoroughly each day.

e. Use the mattress under bag to protect from ground moisture.

2. To keep warm:

a. The mattress is insulated on one side only, therefore, use only with inflating tube on top side.

b. FLUFF BAG thoroughly before using.

c. Wear clean dry winter underwear and socks in the bag. Additional dry clothing such as wool shirt and trousers may be worn in the bag for added warmth.

3. Keep bag CLEAN:

a. ALWAYS wear sleeping hood in bag.

b. Brush and clean clothing before entering bag.

c. Remove dirt and grease from sleeping bag by spot-cleaning with a damp cloth and soap.

4. Slide fastener closure:

To close bag, keep both sides of slide fastener close together before pulling webbing loop on slider. For EMERGENCY EXIT, grasp each side of the opening above the slider and spread apart quickly forcing the slider downward.

NOTE: If chain separates below slider, pull slider down beyond separation, then pull up to reclose chain.

5. Snap fastener closure:

Use the snap fastener flap closure ONLY when the slide fastener fails. Close the bag by snapping the male and female parts together. For EMERGENCY EXIT, grasp each side of the opening above the snap and spread apart quickly.

6. DO NOT smoke in sleeping bag.

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7. Maintenance:

Repair bag in accordance with D/A TM 10-8400-201-23.

3.3.14.3 Identification. The identification label shall conform to type VI, class 5 except as modified in 3.3.14.5 if heat-sealed label is used. The label shall be coated as specified for class 3 labels.

3.3.14.4 Laundry instructions. The laundry instruction labels shall conform to type I, class 3, except as modified in 3.3.14.5 if heat-sealed label is used. The size of the print shall be not less than 3/32 to 1/8 inch in height. All words shown in capital letters shall be 3/16 inch minimum. The laundry instructions shall be as follows:

DOWN AND POLYESTER FILLED

Launder in accordance with formula G of D/A FM 10-17. DO NOT DRY CLEAN.

CAUTION - DRYING TEMPERATURE MUST BE BELOW 130°F.

3.3.14.5 Modification to label requirements for instructions for use and maintenance, laundry instructions and identification (heat-sealed method). The combined label conforming to 3.3.14.2, 3.3.14.3 and 3.3.14.4 shall be modified as required for application to the sleeping bag by heat-sealing. The back of the label shall have a heat activated adhesive coating. The heat-sealed label shall meet the following requirements:

Characteristic	Requirement
Coated weight	6.1 to 8.3 ounces per square yard
Adhesion	5.0 pounds per inch minimum
Hot soap resistance	4.0 pounds per inch minimum
Surface condition and legibility after heat-sealing	1/
Colorfastness	2/

1/ When compared with unsealed labels there shall be no loss in legibility, smearing or bleeding of the print, discoloration, or any loss of the coating on the printed surface.

2/ The label, after heat-sealing, shall show colorfastness to laundering transference and accelerated laundering as specified for class 3 labels.

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3.3.14.6 Content label. If required by state law, a label shall be placed at foot end of bag on inside conforming to the state requirements for the down content used in filling the bag outer panels.

3.4 Design. The design shall be as shown on the drawings listed in 2.1. The sleeping bag shall be of quilted construction with the outside panels filled with down and the inside panels, flap and weatherstrip filled with polyester batting. The front opening shall be equipped with a slide fastener, a flap closure with snap fasteners along the front opening and an adjustable face closure with drawstring.

3.4.1 Batting filled inner panels and weatherstrip. The inner panels consisting of the polyester batting and cotton balloon cloth shall be of quilted construction as shown on Drawing 2-2-328. Spacing of rows of stitching shall be 6 ± 3/4 inches. Location of rows relative to edges (length direction) of panels is optional. Sides and ends of panels shall also be stitched as shown on Drawing 2-2-328. Stitching shall be as specified in operations 8 and 15 of table I. As an option, the weatherstrip filling consisting of the polyester batting may be quilted to the cotton balloon cloth in the same manner as above.

3.5 Patterns. The patterns shall be made by the contractor in accordance with dimensions shown on the applicable drawings. The patterns shall be laid out and cut within plus or minus 1/8 inch of each dimension.

3.6 Construction. The contruction shall conform to the drawings listed in 2.1, table I, and as specified herein.

3.6.1 Stitches, seams and stitching. Stitch, seam and stitching types as specified in table I, as applicable, shall conform to FED-STD-751. When the seam type is not specified, it shall be as shown on the applicable drawing. When more than one method, seam or stitch type is given for the same operation, any one of them may be used. When stitch type 401 is used, the looper thread shall lie between the inner and outer panels of the casing so that the looper thread is not exposed on either the outside or inside panels of the bag. In all sewing operations, the smallest practical needle size shall be used in order to prevent damage to the fabric. Channels shall not be trapped at any seam point so as to prevent a free flow of down throughout all channels on the outer panels.

3.6.1.1 Type 301 stitching. Ends of all stitching shall be backstitched or overstitched 1 inch minimum except where ends are turned under in a hem or held down by other stitching. Thread tension shall be maintained so that there will be no loose stitching resulting in loose bobbin or top thread, or excessively tight stitching resulting in puckering of the materials sewn. The lock shall be embedded in the materials sewn.

3.6.1.1.1 Repairs of type 301 stitching. Repairs of type 301 stitching shall be as follows:

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a. When thread breaks, skipped stitches, or bobbin run-outs occur during sewing, the stitching shall be repaired by restarting the stitching a minimum of 1 inch back of the end of the stitching. 1/

b. Thread breaks, or two or more consecutive skipped or runoff stitches noted during inspection of the item (in-process or end item) shall be repaired by overstitching. The stitching shall start a minimum of 1 inch in back of the defective area, continue over the defective area, and continue a minimum of 1 inch beyond the defective area onto the existing stitching. Loose or excessively tight stitching shall be repaired by removing the defective stitching, without damaging the materials, and restitching in the required manner. 1/

1/ When making the above repairs, the ends of the stitching are not required to be backstitched.

3.6.1.2 Type 401, 502, 503, 504 and 505 stitching. Thread tension shall be maintained so that there will be no loose stitching. All repairs shall be in accordance with 3.6.1.1.1.a. and 3.6.1.1.1.b. Stitching gage for overedge stitching shall be 1/4 + 1/16 inch. Repairs to stitch type 401 may be accomplished by use of stitch type 301.

3.6.1.3 Bartacks. Unless otherwise specified, all bartacks shall be 1/2 + 1/16 inch long and 1/8 + 1/32 inch wide, and shall contain 28 stitches. Bartacking shall be free from thread breaks and loose stitching.

3.6.1.4 Automatic stitching. Automatic machines may be used to perform any of the required stitch patterns provided the requirements for the stitch pattern, stitches per inch, and size and type of thread are met; and at least three tying, overlapping or back stitches are used to secure the ends of the stitching.

3.6.1.5 Thread ends. All thread ends shall be trimmed to a length of not more than 1/4 inch.

3.6.2 Repairs of open seams. Open seams found on quilting or channel stitching or on outer or inner front and back panel joining seams which cannot be repaired in accordance with 3.6.1.1.1 or 3.6.1.2 due to inaccessibility of the seam may be repaired as set forth below.

3.6.2.1 Channel stitching. On the quilting or channel stitching, one repair may be made to each sleeping bag to repair an open seam that is 2 inches or less in length. Using a single strand of 50-2 thread, hand stitch the open seam area using stitch type 202 of FED-STD-751 with 7 to 12 stitches per inch. Stitching shall begin and end a minimum of 1/2 inch beyond the ends of the open seam. At both the beginning and ending of the hand stitching, the ends of the open seam stitching shall be secured by encircling (tacking) the needle and the bobbin (or looper) thread of the seam with two turns of stitching. No more than a 1/8 inch wide bite of the fabric shall be encircled in the stitching ends. Both ends of the thread shall be knotted with an overhand knot firmly tied against the material. The diaphragm on the outer panels shall be caught in the stitching. The outer ply of material on the opposite side of bag shall not be caught in the stitching.

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3.6.2.2 Front and back panel outer and inner joining seams. Along the front and back panel inner or outer joining seam, one repair may be made to each sleeping bag to repair an open seam that is 3 inches or less in length. The open seam shall be repaired using seam type SSc-1, stitch type 301, size 50-2 thread with 10 to 14 stitches per inch. The stitching shall start and stop a minimum of 1 inch beyond the ends of the open seam. Stitch margin shall be 1/16 to 1/8 inch. Ends of stitching shall be secured as specified in 3.6.1.1.

3.6.3 Piecing. There shall be no piecing or splicing of components except as specified herein.

3.6.3.1 Inner panels and weatherstrip. The cotton balloon cloth for the inner panels and weatherstrip may be pieced once only in either the length or width direction. Pieced back and front panels shall be seamed with seam type SSn-1 or LSb-1, with the raw edges on the inside. Width of seam fold shall be 1/2 ± 1/8 inch with stitching 3/16 ± 1/16 inch from edge. Stitch type, stitches per inch and thread size shall be the same as specified in operation 7b of table I.

3.6.3.2 Tape. The 2-1/2 inch tape for the foundation strip may be pieced once in each bag. The 3/4 inch tape for foot straps may be pieced so that the seam will not be visible on the outside of the bag. Piecing of 3/4 inch tape shall be accomplished by overlapping the ends to be joined 1 inch with each raw edge turned under 1/2 inch and with the overlap boxstitched. Piecing of the 2-1/2 inch tape shall be accomplished by overlapping 1/2 inch of bottom piece and 1 inch of top piece with the top piece turned under 1/2 inch and the overlap boxstitched. Tolerance for dimensions specified above shall be ± 1/8 inch. Stitch type for piecing tape shall be 301. Stitches per inch and thread size shall be as specified in operation 3b of table I. Stitch margin shall be 1/8 ± 1/16 inch.

* 3.6.3.3 Flap. The cotton and nylon oxford cloth for the flap cover may be pieced once in the lengthwise direction as shown in the alternate construction on Drawing 2-2-362. Stitch type, stitches per inch, and thread size shall be as specified in operation 4 of table I.

3.6.4 Fusing of nylon lace ends. All ends of nylon lace shall be fused. The apparatus used to fuse the lace ends shall be capable of providing sufficient heat to provide a smooth edge with the cut ends of the lace yarns all fused together.

3.6.5 Setting of snap fasteners. A hole shall be prepunched through the materials before inserting the eyelet component of the snap fastener. The hole punched before inserting the eyelet component shall be smaller than the outside diameter of the fastener barrel so that the barrel must be forced through the hole. The fasteners shall be securely clinched without cutting the adjacent material and no more than three splits shall occur in the button or eyelet barrels.

3.6.6 Cutting of non-woven cloth. In the cutting of the non-woven cloth (see operation "1a" of table I) no spliced, open end or lapped diaphragm material shall be allowed to be utilized in the cut parts.

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3.6.7 Repairs. Repairing the sleeping bag by mending, patching or darning is not allowed.

3.6.8 Replacement of defective components. During the spreading, cutting and manufacturing process, components having material defects or damages that are classified as defects in 4.4.3 shall be removed from production and replaced with non-defective and properly matched components.

3.7 Manufacturing operations requirements. The sleeping bag shall be manufactured in accordance with the operation requirements specified in table I, as applicable. The contractor is not required to follow the exact sequence of operations unless otherwise specified.

* 3.8 Weighing of unfilled and filled casings. For construction methods A & B the weighing of unfilled and filled casings shall be conducted on a bench dial indicating scale conforming to type I of AAA-S-108 except the size shall be 20 pound maximum capacity with minimum load receiver dimensions of 20 inches by 18 inches. The maximum graduation or resolution shall be 1/4 of an ounce increments. The scale shall be direct reading without any loose weights added. For construction method C the weighing of unfilled and filled front and back panels shall be conducted on a electronic type scale with readouts to 0.01 lb. As an option the electronic scale may be used for construction methods A or B.

3.9 Location marks. Location marks shall not be drilled except for marking location of the instruction/care label. Drill holes for this purpose shall be covered by the label otherwise exposed drill holes shall be considered to be holes in the fabric.

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TABLE I.

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
	Construction method "A"						
1.	Cutting.						
	a. The inner and outer panels and diaphragms for outer panels shall be cut in strict accordance with the measurements shown on the applicable drawings. The diaphragms shall be cut the same size as the outer panels. No spliced, open end or lapped diaphragm material shall be allowed to be utilized in the cut parts.						
	b. All parts shall be cut on the straight of the material, that is, in the warp direction, except that the weatherstrip balloon cloth may be cut on the straight or across the material.						
2.	Apply markings and labels.						
	a. Apply marking "US" and the type of sleeping bag at foot end of bag as specified in 3.3.14.1. The type shall be applied to both sides of the bag.						
	b. Place the combined label for use and maintenance, laundry instructions and identification in the location shown on Drawing 2-2-328 and attach to inner back panel. If stitched method is used, stitch on all four sides 1/16 to 1/8 inch from edge.	301	-	8-10	50-2	50-2	
	If heat-sealed method is used, apply with a heat-sealing machine designed for patch and label application and equipped for automatic regulation of temperature, dwell time, and dwell pressure. The maximum temperature allowed for heat-sealing the label is 390°F. Application by hand iron is not permissible. The component of the heat-sealing						

NATICK Form 903
1 Dec 76 EDITION OF 1 OCT 76 WILL BE USED UNTIL EXHAUSTED.

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TABLE I. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
2.	Apply markings and labels. (cont'd)						
	apparatus coming in contact with the surface of the label shall be cleaned whenever necessary to prevent smearing of the label. The label shall read from head end of bag to foot end.						
*3.	Fill outerfront and back panels with down.						
	a. Temporarily secure left and right outer front panels together by pining or stitching. (Method used shall assure that the panels do not separate prior to being assembled to inner panels).						
	b. Record in ink or indelible pencil the weight of the joined front panels and the back panel prior to filling with down. The weight shall be recorded on an attached commercial type tag or the weight may be stamped directly on the panels at the foot end using a black marking medium. In addition to the weight the marking "Construction C" shall be included. The printing shall be legible and shall not show offsetting, smearing or bleeding. Height of characters shall be 3/16 inch minimum (see 3.3.14.1).	1/2 inch bartack or 301	EFb-1	28 per bartack 8-10	50-2 50-2	50-2 50-2	
	c. Fill the joined front panels with 0.94 pounds of down minimum and fill the back panel with 1.93 pounds of down minimum. The filling procedures shall be as specified in operation 26b, table I.						
4.	Make flap assembly.						
	Make flap assembly in accordance with Drawing 2-2-362. The stitching shall by type 301 except type 401 may be used for the row of quilting stitching.	301 401	- -	8-12 8-12	50-2 50-2	50-2 70-2	

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TABLE I. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
5.	Make weatherstrip.						
	The weatherstrip shall be made in accordance with Drawing 2-2-329. Stitching shall conform to type 301 except tape may be sewn on with type 301 or 401.	301 or 401	-	8-12 8-12	50-2 50-2	50-2 70-2	
6.	Make slide fastener - weatherstrip assembly.						
	The slide fastener - weatherstrip assembly shall be made in accordance with Drawing 2-2-329 or the optional method shown in optional section B-B on Drawing 2-2-330.	301	-	8-10	30-3	30-3	
7.	Make channels for the outer down filled front panels (typical left and right sides).						
	a. The outer front panel shall be made of one ply of cotton/nylon oxford cloth which will be the sleeping bag casing outer ply and a ply of one of the materials specified in 3.3.2 for the diaphragm (see 3.6.6). The diaphragm shall finish even with the ends and sides of the casing outer ply after completion of the channel stitching ± 1/4 inch.						
	b. Form the channels by stitching the diaphragm to the casing outer ply as shown on Drawing 2-2-328.	301 or 401	SSa-1 SSa-1	10-14 10-14	50-2 50-2	50-2 70-2	
8.	Make batting/balloon cloth quilted front inner panels (typical left and right sides).						
	a. The inner front panel shall be made of one ply of cotton balloon cloth, which will be the sleeping bag casing inner ply, and a layer of polyester batting.	301 or 401	SSa-1 SSa-1	6-10 6-10	50-2 50-2	50-2 70-2	

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TABLE I. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
8.	Make batting/balloon cloth quilted front inner panels (typical left and right sides). (cont'd)						
	The batting shall be the same size as the inner ply and shall be quilted to the balloon cloth as shown on Drawing 2-2-328 and as specified in 3.4.1.						
	b. Stitch batting to balloon cloth along three edges as shown on Drawing 2-2-328.	301 or 401 or 502, 503, 504 or 505	SSa-1 SSa-1 SSa-1	6-10 6-10 6-10	50-2 50-2 70-2	50-2 70-2 70-2	
9.	Join inner and outer front panels (typical left and right sides).						
	Join the inner and outer front panels with row of stitching along the front opening using one of the two methods shown on Drawing 2-2-328.	301 or 401	SSa-1 SSa-1	8-10 8-10	50-2 50-2	50-2 70-2	
10.	Form front panel sub-assembly by joining left and right front panels.						
	a. Starting at the foot end of the panels and proceeding up to the notch, or starting at the notch and proceeding to the foot end of the panels, stitch inner plies of both panels together as shown on Drawing 2-2-330.	301 or 401 or 502, 503, 504, 505	SSa-1 SSa-1 SSa-1	10-14 10-14 6-10	50-2 50-2 70-2	50-2 70-2 70-2	
	b. Starting at the foot end of the panels and proceeding up to the notch, or starting at the notch and proceeding to the foot end of the panels, stitch outer plies of both panels together catching the diaphragms of each panel in the seam as shown on Drawing 2-2-330.	301 or 401	SSa-1 SSa-1	10-14 10-14	50-2 50-2	50-2 50-2	

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TABLE I. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN / LOOPER	COVER
11.	Attach reinforcement tape to notch of front panel sub-assembly.						
	Stitch a 3/4 inch wide reinforcement tape to the inside of the front panel assembly to reinforce the notch. Size, location and stitching shall be as shown on Drawing 2-2-330.	301	-	8-10	30-3	30-3	
12.	Attach slide fastener - weatherstrip assembly to front panel sub-assembly.						
	a. Position the edges of the front panels between the slide fastener and foundation strips of the slide fastener - weatherstrip assembly as shown on Drawing 2-2-330 and stitch with two rows of stitching on each side, catching the front panels with both rows of stitching as shown on Drawing 2-2-330. The weatherstrip balloon cloth shall not be caught in the stitching.	301	SSa-2	8-10	30-3	30-3	
	b. At the foot end, bartack in three places as specified on Drawing 2-2-330.	1/2 inch bartack	-	28 per bartack	30-3	30-3	
13.	Attach male snap fastener.						
	a. Attach male snap fasteners on the slide fastener guard strip as shown on Drawing 2-2-330.						
	b. Attach each snap fastener retainer to front panel sub-assembly with two 1/2 inch bartacks as specified on Drawing 2-2-330.	1/2 inch bartack	-	28 per bartack	30-3	30-3	

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NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
14.	Make channels for outer down filled back panel.						
	a. The outer back panel shall be made of one ply of the cotton/nylon oxford cloth which will be the sleeping bag casing outer ply, and a ply of one of the materials specified in 3.3.2 for the diaphragm (see 3.6.6). The diaphragm shall finish even with the ends and sides of the casing outer ply after completion of the channel stitching, ± 1/4 inch.						
	b. Form longitudinal channels by stitching the diaphragm to the casing outer ply as shown on Drawing 2-2-328.	301 or 401	SSa-1 SSa-1	10-14 10-14	50-2 50-2	50-2 70-2	
	c. At the foot end of the back panel cut diaphragm at each side of each row of channel stitching back 3/4 ± 1/4 inch from the edge of the cotton/nylon oxford cloth.						
15.	Make batting/balloon cloth quilted inner back panel.						
	a. The inner back panel shall be made of one ply of cotton balloon cloth, which will be the sleeping bag causing inner ply, and a layer of polyester batting. The batting shall be the same size as the inner ply and shall be quilted to the balloon cloth as shown on Drawing 2-2-328 and as specified in 3.4.1	301 or 401	SSa-1 SSa-1	6-10 6-10	50-2 50-2	50-2 70-2	
	b. The edges along the sides and foot end of the panel shall be stitched as shown on Drawing 2-2-328.	301 or 401 or 502, 503 504 or 505	SSa-1 SSa-1 SSa-1	6-10 6-10 6-10	50-2 50-2 70-2	50-2 70-2 70-2	

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NO.	TABLE I. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
16.	Seam front panels to back panels.						
	a. At the shoulder end of the outer front panel (down filled) cut diaphragm at each side of the two rows of channel stitching back 3/4 ± 1/4 inch from the edge of the cotton/nylon oxford cloth outer ply.						
	b. Stitch outer ply of each front panel to the outer ply of the back panel as shown on Drawing 2-2-325, starting at the top end forming the shoulder seam. The diaphragm at the shoulder seam end shall be folded back the distance cut back in operation 16a above. Continue stitching the full length of the sides of the panels. The diaphragms of the back and front panels shall be caught in the seam along the sides but not in the shoulder seam area.	301 or 401	SSa-1 SSa-1	8-10 8-10	30-3 30-3	30-3 50-2	
	c. Stitch inner ply of each front panel to the inner ply of the back panel the full length of the sides as shown on Drawing 2-2-325. Leave shoulder seam open. The batting shall be fully caught along the length of the seam.	301	SSa-1	8-10	30-3	30-3	
	d. To provide clearance for the face opening binding tape (see operation 24) in the shoulder seam, stitch the inner ply of each front panel to the inner ply of the back panel for a distance of 1 inch minimum. Leave remainder of seam open to facilitate filling of outer front panels with down.	301	SSa-1	8-10	30-3	30-3	

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TABLE I. (cont'd)

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					NEEDLE	BOBBIN/ LOOPER	COVER
17.	Form head end of bag.						
	a. Close outer ply of casing at head end by joining raw edges and stitching as shown on Drawing 2-2-325. The edge of the diaphragm shall be closed and caught in the seam.	301	SSa-1	10-14	50-2	50-2	
	b. Close inner ply of casing at head end by joining raw edges and stitching as shown on Drawing 2-2-325. The batting shall be fully caught along the length of the seam.	301	SSa-1	10-14	50-2	50-2	
18.	Seam foot end.						
	a. Seam outer plies of casing at the foot end with the foot straps located and caught in the seam as shown on Drawing 2-2-325. The diaphragm shall be folded back the distance cut back in operation 14c. and shall not be caught in the stitching. The content label when required, shall also be caught in the seam between the foot straps.	301 or 401	SSa-1 SSa-1	8-10 8-10	30-3 30-3	30-3 50-2	
	b. Bartack each foot strap through outer plies of casing as shown on Drawing 2-2-325.	1/2 inch bartack	-	28 per bartack	50-2	50-2	
19.	Trim face opening.						
	Trim the top ends (left and right sides) of weather-strip assembly at approximately a 30° angle parallel to the face opening.						

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					NEEDLE	BOBBIN/ LOOPER	COVER
20.	Seam edges of face opening (optional).						
	As an option, edges of face opening may be seamed as shown on Drawing 2-2-325. Head seams of outer and inner plies of the casing shall finish seam on seam within 1/4 inch.	301 or 401	SSa-1 SSa-1	6-10 6-10	50-2 50-2	50-2 70-2	
21.	Make face closure drawstring casing.						
	a. Cut drawstring casing of the required length and width, as shown on Drawing 2-2-325, from cotton/nylon oxford cloth. Double fold each end of casing and stitch as shown in section A-A, Drawing 2-2-325.	301	EFb-1	8-10	50-2	50-2	
	b. Evenly fold casing in half in length direction. At each end, bartack through double folded hem with a 1/2 inch bartack positioned parallel with length of casing and located 7/16 $\pm$ 1/16 inch from the outer folded side edge and (1/8 $\pm$ 1/16 inch) from the end of the casing.	1/2 inch bartack	-	28 per bartack	50-2	50-2	
22.	Make and install drawstrings in casing.						
	a. Cut two drawstrings 30 $\pm$ 1/2 inches long from nylon lace specified in 3.3.5 and fuse cut ends (see 3.6.4). Knot one end of each drawstring with a figure eight as specified on Drawing 2-2-325.						
	b. At one end of casing, pass unknotted end of one drawstring through opening formed by bartack and position drawstring within the casing with the unknotted end protruding approximately two inches from other end of casing and on opposite side of						

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					NEEDLE	BOBBIN/ LOOPER	COVER
22.	Make and install drawstrings in casing. (cont'd) bartack, i.e., not within the opening formed by the bartack. Install other drawstring in the same manner except start from other end of casing.						
23.	Attach face closure drawstring casing to face opening.						
	a. With drawstrings in proper position within the casing and casing evenly folded along its length, position casing on inside surface of face opening with the two aligned edges positioned even with the edge of the face opening and the ends located even with the outer edge of the slide fastener tape, on the left side ± 1/4 inch, and shall be positioned on the right side 1/2 inch from edge of fastener tape ± 1/4 inch. Stitch casing to face opening with row of stitching located 3/8 ± 1/16 inch from edge of face opening. Head seams of outer and inner plies of the casing shall finish seam on seam within 1/4 inch.	301	SSa-1	8-10	30-3	30-3	
	b. On right side of face opening, fold the end of drawstring that is unknotted under 3/8 to 1/2 inch, and position folded edge even with end of 1-1/4 inch wide slide fastener foundation strip and abutted against the end of the drawstring casing. Bartack through the folded end of the drawstring and slide fastener guard strip with a 1/2 inch bartack running parallel with length of drawstring.	1/2 inch bartack	-	28 per bartack	30-3	30-3	

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					NEEDLE	BOBBIN/ LOOPER	COVER
23.	Attach face closure drawstring casing to face opening. (cont'd)						
	c. Secure unknotted end of drawstring on left side of face opening in same manner as in operation 23b above, with folded edge even with end of 1-1/4 inch wide slide fastener left guard strip and with bartack passing through weatherstrip.						
24.	Bind face opening.						
	a. Bind the face opening with 1-1/4 inch tape as shown on Drawing 2-2-325. Binding shall also encompass the ends of the weatherstrip, including foundation strip, and the slide fastener guard strip on the left side and on the right side the end of the slide fastener foundation strip.	301	BSa-2	8-10	30-3	30-3	
	b. Ends of binding tape shall be turned under and box stitched as shown on Drawing 2-2-325.	301	EFa-1	8-10	30-3	30-3	
25.	Inspect casing for general construction.						
26.	Fill bag						
	a. Record in ink or indelible pencil, the weight of the casing, prior to filling with down, on an attached commercial type tag; or the weight may be stamped on the inside of the bag at the foot end using a black marking medium. In addition to the weight the markings "Construction A" shall be included on the tag or with the direct printing. The printing shall be legible and shall not show offsetting, smearing or bleeding. Height of characters shall be 3/16 inch minimum. (See 3.14.1).						

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					NEEDLE	BOBBIN/ LOOPER	COVER
26.	Fill bag. (cont'd)						
	b. The apparatus and procedure for filling the bag shall be such that the down is evenly distributed thru each individual channel (i.e., each channel shall be evenly filled along the entire length of the channel). Each channel shall have its proportionate amount of down, i.e. as compared to other channels.						
	c. Fill outer channels of the bag with not less than 46 ounces of down (see operation 29).						
27.	Close bag.						
	a. At the foot end of the bag, turn the edges of the diaphragm that were folded back in operation 18a. back to the edge of the outer plies and stitch through diaphragm and outer plies as shown on Drawing 2-2-325.	301 or 401	SSa-1 SSa-1	8-10 8-10	30-3 30-3	30-3 50-2	
	b. Turn the edges of the inner ply at the foot end of the bag to the inside and double stitch, closing the foot end of the bag and catching the footstraps in the stitching as shown on Drawing 2-2-325. The exposed portion of the loop of each footstrap and the distance between the stitching of the outer ply and inner ply of the casing shall be as shown on Drawing 2-2-325.	301	-	8-10	30-3	30-3	

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					NEEDLE	BOBBIN/ LOOPER	COVER
27.	Close bag. (cont'd)						
	c. At the shoulder seams, turn the edges of the diaphragm that were folded back in operation 16b back to the edge of the outer plies and stitch through diaphragm and outer plies as shown on Drawing 2-2-325.	301 or 401	SSa-1 SSa-1	8-10 8-10	30-3 30-3	30-3 50-2	
	d. Turn the raw edges at shoulders and seam, closing both shoulders of inner ply with two rows of stitching as shown on Drawing 2-2-325. Batting shall be fully tucked into the seam, i.e., no batting shall be exposed.	301	SSc-2	8-10	30-3	30-3	
28.	Clean bags.						
	Trim all loose ends of the thread to a length of not more than 1/4 inch and remove from bag. Remove all spots and stains. The finished bag shall be cleaned of all clinging down on the outside and inside surfaces.						
29.	Recording						
	a. Record weight of filled bag as in operation 26a (see 3.8 and operation 26c).						
	b. Contractor shall not remove weight tag (if used in lieu of direct printing).						

TABLE I. (cont'

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					NEEDLE	BOBBIN/ LOOPER	COVER
	Construction method "B"						
1.	Perform operations 1 through 16b, method A.						
2.	Perform operation 17a, method A.						
3.	Stitch around edge of outer panel face opening.						
	Stitch diaphragm to outer panel around entire face opening with row of stitching located 1/4 ± 1/16 inch from edge of outer panel.	301 or 401	SSa-1 SSa-1	10-14 10-14	50-2 50-2	50-2 70-2	
4.	Perform operations 18a and 18b, method A.						
5.	Perform operations 25 and 26, method A.						
6.	Perform operations 27a and 27c, method A.						
7.	Record weight of casing after filling. Record the weight of the casing after filling outer panels with down. Marking shall be in accordance with operation 26, method A.						
8.	Seam inner back panel to inner front panels.						
	Stitch the inner ply of each front panel to the inner ply of the back panel the full length of the sides and shoulders as shown on Drawing 2-2-325. The batting shall be fully caught along the length of the seams.	301	SSa-1	10-14	50-2	50-2	
9.	Perform operation 27b, method A.						

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NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
10.	Perform operation 17b, method A.						
11.	Perform operations 19 through 25 and 28, method A. (Contractor shall not remove weight tag if used in lieu of direct printing.)						

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					NEEDLE	BOBBIN/ LOOPER	COVER
	Construction method "C"						
1.	Perform operations 1 through 7b, method A except add the following to operation 7b: "In addition, stitch diaphragm to panel along the side and diagonal edge leaving top end open only. Stitch margin shall be 1/4 + 1/16, - 1/8 inch."						
2.	Perform operations 14a and 14b, method A except add the following to operation 14b: "Stitch diaphragm to panel along the side edges and the head end leaving foot end open only. Stitch margin shall be 1/4 + 1/16, - 1/8 inch."						
3.	Join outer front panels to outer back panel and fill with feathers and down.						
	a. Temporarily secure left and right outer front panels to outer back panel by pining or stitching. (Method used shall assure that the panels do not separate prior to being assembled to inner panels).						
	b. Record in ink or indelible pencil, the weight of joined panels, prior to filling with down, on an attached commercial type tag; or the weight may be stamped directly on the back panel at the foot end using a black marking medium. In addition to the weight the marking "Construction C" shall be included on the tage or with the direct printing. The printing shall be legible and shall not show offsetting, smearing or bleeding. Height of characters shall be 3/16 inch minimum. (See 3.3.14.1).						

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					NEEDLE	BOBBIN/ LOOPER	COVER
3.	Join outer front panels to outer back panel and fill with feathers and down. (cont'd)						
	c. Fill outer front and back panels per operations 26b and 26c, method A.						
	d. Close open ends of panels with single row of stitching located 1/4 ± 1/16 inch from edge of panels.	301 or 401	SSa-1 SSa-1	10-14 10-14	50-2 50-2	50-2 70-2	
	e. Record weight of the filled panels as in operation b above. (Contractor shall not remove weight tag if used.)						
4.	Perform operations 8 through 13, and operation 15, method A.						
5.	Seam front panels to back panels.						
	Stitch outer and inner front panels to the outer and inner back panels as shown on Drawing 2-2-325, section F-F and construction "B" of section B-B, starting at the top ends forming the shoulder seams and continuing the full length of the sides of the panels. When joining the inner panels the batting shall be fully caught along the length of the seam.	301 or 401	SSa-1 SSa-1	8-10 8-10	30-3 30-3	30-3 50-2	
6.	Perform operation 17, method A.						
7.	Seam foot end.						
	a. Seam outer plies of casing as specified in operation 18a, method A except the second sentence shall not apply and the stitching shall pass through the diaphragm.						

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TABLE I. (cont'd)

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					NEEDLE	BOBBIN/ LOOPER	COVER
7.	Seam foot end. (cont'd)						
	b. Bartack footstraps as specified in operation 18b, method A except the bartack shall pass through the diaphragm.						
	c. Seam inner plies of casing as specified in operation 27b, method A.						
8.	Perform operations 19 through 25, and 28, method A.						

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3.10 Workmanship. The finished sleeping bags shall conform to the quality of product established by this document. The occurance of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects specified in 4.4.3, 4.4.4 and 4.4.5. The presence of any defect shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all requirements of referenced documents unless otherwise excluded, amended, modified or qualified, in this document or applicable purchase document. In addition, testing shall be performed on the heat-sealed label (when used) as specified in table II. The lot size shall be expressed in terms of labels. The inspection level shall be S-1. Unless otherwise specified, the lot shall be unacceptable if one or more sample units fail to meet any test requirement specified.

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TABLE II. Component testing of labels (heat-sealing)

Characteristic	Requirement paragraph	Test procedure	Requirements applicable to	No. deter./ sample unit	Results reported to	Sample unit
Coated weight	3.3.14.5	5041 1/	Sample unit	1	Nearest 0.1 oz/ sq. yd.	5 labels
Adhesion	3.3.14.5	4.5.1	Sample unit	1	Nearest 0.1 lb.	5 specimens
Hot soap resistance	3.3.14.5	4.5.1	Sample unit	1	Nearest 0.1 lb.	5 specimens
Colorfastness	3.3.14.5	2/	-	-	-	-

1/ Test method of FED-STD-191.

2/ Labels shall be tested for colorfastness to laundering transference and accelerated laundering as specified in DDD-L-20 for class 3 labels except the labels shall consist of specimens which have been heat-sealed to cotton cloth conforming to 3.3.1 and which were heat-sealed in the same manner as done for the preparation of the specimens for the adhesion and hot soap resistance tests.

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4.4.1.1 Certification. The contractor shall furnish a certificate of compliance for the non-woven cloth requirement specified in 3.3.2.

4.4.2 In-process inspection. Inspection of sub-assemblies shall be made to ascertain that construction details which cannot be examined in the finished product are in accordance with specified requirements. This examination shall include inspection of the inner front and back panels to assure that the batting is caught in the stitching along the entire length of the side edges and also inspection of the front and back panel joining seams to assure that the correct seam allowance is being used. Whenever non-conformance is noted, corrections shall be made to the parts affected and lot in-process. Parts which cannot be corrected shall be removed from production.

4.4.2.1 Inspection of procedure for filling outer panels, construction method A, B or C. Inspection shall be performed to determine that the procedure required in operation 26 of table I for filling the outer panels, construction A, B, or C is performed as specified. Outer panels not filled as specified shall be considered not acceptable (see channel examination under 4.4.3).

4.4.2.2 Testing of heat sealed labels. Testing shall be performed on the heat-sealed label (when used) in accordance with 4.5.2 to determine conformance to the label adhesion requirement specified in 3.3.14.5. The lot size shall be expressed in terms of inner back panels (sewed or unsewed) or unfilled casings. The sample unit shall be sufficient inner back panels or unfilled casings to prepare the required number of test specimens (see 4.5.2.1). The inspection level shall be S-1. One determination shall be made per sample unit and the results shall be reported to the nearest 0.1 pound. Failure of any sample unit to meet the requirement specified shall be cause for rejection of the lot.

4.4.2.3 Inspection for weight of unfilled casings and outer panels. Inspection of unfilled casings, construction method A or B and unfilled outer panel assembly (joined front and back outer panels), construction method C shall be made to verify the weight recorded by the contractor. The sample unit shall be one unfilled casing or one unfilled outer panel assembly as applicable. The defects found during examination shall be classified in accordance with the list below. The inspection level shall be I and the acceptable quality level (AQL), expressed in terms of defects of hundred units, shall be 1.0 for major defects and 4.0 for major and minor defects combined.

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Examine	Defect	Classification Major	Minor
	NOTE: Utmost care shall be taken to assure that the apparatus used to perform weight examination is properly calibrated and accurate (see 3.8).		
Weight of unfilled casing construction methods A&B	In excess of weight indicated:		
	- by 1/2 ounce up to one ounce		X
	- by more than one ounce	X	
Weight of joined outer front panels, or outer back panel, construction method C	Joined outer front panels in excess of weight recorded:		
	- by 0.01 pound up to 0.02 pound		X
	- by more than 0.02 pound	X	
	Back panel in excess of weight recorded:		
	- by 0.02 pound up to 0.04 pound		X
	- by more than 0.04 pound	X	

* 4.4.2.4 Inspection for weight of filled casing, construction method B and filled outer panels, construction method C. Inspection of filled casings, construction method B and unfilled outer panels, construction method C shall be made to verify the weight of the filled casing or outer panels as applicable. For construction method B the sample unit shall be one filled and closed casing with the foot and shoulder seams of the outer front and back panels closed. For construction method C the sample unit shall be one filled and closed outer back panel or one joined, filled and closed outer front panel assemble, as applicable. Defects found during examination shall be classified in accordance with the list below. The inspection level shall be I and the AQL, expressed in terms of defects per hundred units, shall be 1.0 for major defects and 4.0 for major and minor defects combined.

Examine	Defect	Classification Major	Minor
Weight of filled casing, construction method B 1/	Under the required total weight of down by 1 ounce or more	X	
	Under the required total weight of down by 1/2 ounce up to but not including 1 ounce		X

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Examine	Defect	Classification Major	Minor
Weight of joined outer front panel or outer back panel after filling, construction method C 1/	Joined outer front panels under the required total weight of down: - by 0.02 pound or more	X	
	- by 0.01 pound up to but not including 0.02 pound		X
	Back panel under the required total weight of down:		

1/ The weight of the down shall be as specified in the applicable operation of table I. To determine the weight of the down in the casing, construction method B or the outer panels, construction method C, deduct the recorded weight the recorded weight of the unfilled casing or outer panel assembly from the weight of the filled casing or outer panel assembly. Utmost case shall be taken to assure that the apparatus used to perform the weight examination is properly calibrated and accurate (see 3.8).

4.4.2.5 Inspection of down fill. Random samples of down shall be taken from the filling tubes to determine that the quality of the down used in the sleeping bags conforms to the requirements of 3.3.10. The sample unit quantity and sample size shall be as stated in MIL-F-43907 for the weight of down used in the sleeping bag lot. Failure of any test sample to be within the specified limits for down characteristics shall be cause for rejection of the inspection lot.

4.4.3 End item visual examination. The end item shall be examined for defects listed below. The sample unit shall be one finished sleeping bag. The lot size shall be expressed in units of one sleeping bag. The inspection level shall be II and the AQL, expressed in terms of defects per hundred units, shall be 2.5 for major defects and 25.0 for total (major and minor combined) defects.

Examine	Defect	Classification Major	Minor
Fabric	Hole, cut or tear		X
	Any abrasion mark, smash, large slub, broken or missing yarn, multiple float, or open place, clearly visible at normal inspection distance (3 feet).	X	
	Fine or coarse filling bar.		X
	Any mend, darn, patch or unauthorized splice.	X	
	Needle chews.	X	

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Examine		Defect	Classification Major	Minor
Fabric (cont'd)	NOTE:	Needle holes visible as the result of broken or skipped stitching or stitching that has been removed shall not be considered as needle chews providing that the holes are spaced as in normal stitching.		
Webbing and tape		Cut or frayed edge.	X	
		Not firmly and tightly woven.		X
		Needle chews:		
		- up to 1/8 inch in length		X
		- over 1/8 inch in length	X	
Open seams:	NOTE:	A seam shall be classified as open when one or more stitches joining a seam are broken or when two or more consecutive skipped or runoff stitches occur. On double-stitched seams, a seam shall be considered open when either one or both sides of the seam are open.		
On channel stitching (applicable inner and outer panels - individual seams)		More than 1/2 inch		X
On seams that would allow escape of down from the bag		Any open seam	X	
On all seams other than above		1/2 inch or less		X
		More than 1/2 inch	X	
		Any open seam on both rows of double stitched seams.	X	
Raw edges (when securely caught in stitching):				
Along face and front opening		1/2 inch or more.		X

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Examine	Defect	Classification Major	Classification Minor
Raw edges (when securely caught in stitching) (cont'd)			
On all other parts of bag	One inch or more.		X
NOTE:	Raw edges not securely caught in stitching shall be classified as open seams.		
Seam and stitch type	Wrong seam or stitch type.	X	
Stitch tension	Loose tension, resulting in a loose top or bottom thread.		X
	Excessively tight tension resulting in puckering of material.		X
NOTE:	Defects to be scored only when the condition exists for 4 inches or more or in several areas with an accumulated distance of 8 inches or more. Applicable to individual seams.		
Stitches per inch	One to two less than minimum specified.		X
	Three or more less than minimum specified.	X	
	One or more in excess of maximum specified.		X
NOTES:	(1) Variation in the number of stitches per inch caused by operator speeding up the machine and pulling the material in order to sew over heavy places, or in turning corners shall be classified as follows: (a) Within the minor defect classification - no defect (b) Within the major defect classification - minor defect		
	(2) Defect to be scored only when condition exists on any one seam for a length of 6 inches or more or when the combined length of several areas exceeds 10 inches		

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Examine	Defect	Classification Major	Minor
Rows of stitching	Any row omitted (unless otherwise classified herein).	X	
Stitching for joining front panels to slide fastener - weatherstrip assembly	On inside of the bag, one row of stitching runoff slide fastener - weatherstrip assembly:		
	- for a distance of 3 inches up to 8 inches or in several areas with an accumulated distance of 8 to 12 inches		X
	- for a distance of more than 8 inches or in several areas with an accumulated distance of more than 12 inches.	X	
	On inside of bag, both rows of stitching runoff slide fastener - weatherstrip assembly; or on outside of bag, one or both rows of stitching runoff slide fastener - weatherstrip assembly. (Classify in accordance with "Thread breaks, skipped stitches or runoffs" defect.)		
Stitching ends	Ends of stitching not secured as specified (except when caught in other seams or stitching).		X
Thread breaks, skipped stitches, or runoffs (unless otherwise classified herein)	Overstitched less than 1 inch in each direction beyond the defective stitching area.		X
	NOTE: Thread breaks or two or more consecutive skipped or runoff stitches not overstitched shall be classified as open seams.		

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Examine	Defect	Classification Major	Minor
Bartacks	Stitching loose, incomplete, or broken.		X
	Any bartack omitted.	X	
Components and assembly	Any component part omitted or not as specified or required operation omitted (unless otherwise classified).	X	
	Any seam badly puckered, twisted, or pleated.		X
	Repairs of open seams allowed by 3.6.2:		
	- not performed as specified		X
	- more than one of either type made per sleeping bag. (Applicable to each repair over amount allowed)		X
	Any unauthorized repair.	X	
	Any unrelated part of bag caught in a seam.	X	
	Material for inner panel used for outer panel or vice versa.	X	
	Batting filled panels on outside of bag in lieu of inside.	X	
Piecing:			
Flap	Cover of flap is constructed from more than two pieces		X
	Location and method of piecing not as specified		X
Casing and weatherstrip	Any cloth component part of casing or weatherstrip constructed with more than two parts.		X
Foundation strip (2-1/2 inch tape) and foot straps	Strip or straps constructed with more than two pieces.		X
	Location and method of piecing not as specified.		X
Channels	Number of channels less than specified.	X	
	Any channel on outer panels trapped, causing part of channel to remain unfilled with down.	X	
	Badly twisted between rows of channel stitching.		X

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Examine	Defect	Classification Major	Minor
Channels (cont'd)	Any channel improperly filled.	X	
NOTE:	An improperly filled channel is one that meets any one of the following conditions: a. No down along entire length or no down in a portion of the channel. b. Less than its required proportionate amount of down compared to other channels. c. One in which the down is not equally distributed throughout the entire length of the channel.		
Instruction and identification labels	Omitted, misplaced, data not as specified, illegible.		X
	Any area around periphery of label not adhered exceeding 1 inch in length and more than 1/8 inch in from edge of label (applicable to heat-sealed label).	X	
	Smearing or bleeding of print, discoloration, or loss of coating on printed surface.		X
Content label (if required)	Omitted, illegible, incorrect or misplaced.		X
Markings "U.S." and Extreme Cold Weather	Omitted, illegible, misplaced or not specified type.		X
Tag for indicating construction method, weight of casing or panel assembly and total weight of bag	Omitted, data illegible or omitted, construction method omitted.	X	

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Examine	Defect	Classification Major	Minor
Stamped weight and construction method	Printing illegible, smeared or bleeding, data omitted, construction method omitted, size of characters not as specified.		X
Cleanness	Spots or stains, clearly noticeable; clinging down not removed throughout bag, thread ends not trimmed as specified.		X
Face opening	Binding twisted or pleated causing excessive puckering or pleats at face opening.		X
	Ends of binding not finished as specified.		X
	Free ends of drawstrings not knotted or any end not seared.		X
	Ends of drawstrings not secured as specified.		X
Snap fasteners, general	Not specified style.	X	
	Finish omitted or not as specified.		X
	One or more omitted, broken or malformed failing to serve intended purpose; burrs, or sharp edges.	X	
	Insecurely clinched to a degree that one or more fasteners may be detached from flap or tape; one or more excessively clinched cutting adjacent material.		X
	Four or more splits in button or eyelet barrel.		X
	Edge of male snap fastener projects more than 1/16 inch beyond edge of webbing.	X	
Slide fastener	Not specified type or size.	X	
	Does not provide a smooth and secure closure full length of bag opening.	X	
	Slider jams or fails to interlock chain scoops.	X	
	Color of components not as specified.		X
	Fastener tape width not as specified.	X	

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Examine	Defect	Classification Major	Minor
Slide fastener (cont'd)	Any portion of fastener broken, bent, missing or not aligned making fastener unusable.	X	
	Fastener tape not 1 inch wide.		X
	Slider not double pull, not furnished with two stirrup pulls or stirrup pulls not made out of wire.		X
	Thong(s) omitted.	X	
	Chain not plastic or not of the configuration specified.	X	
	NOTE: The slide fastener shall be fully closed and opened to determine whether fastener operates smoothly and provides a secure closure.		
Slide fastener - weatherstrip and flap assembly	Stitching too close to chain, preventing slider from sliding freely.		X
	Top end of fastener guard strip not fully and securely caught in face opening binding.		X
	End at bottom end not turned under.		X
	Ends of foundation strip at bottom end of fastener assembly not folded and stitched.		X
	Bartacks at bottom of assembly do not pass through reinforcement tape.		X
	Weatherstrip placed on right side in lieu of left side of front opening.		X
	Top end of weatherstrip not fully caught in binding of face opening.		X
	Batting not tucked in when closing weatherstrip or flap along side and ends for a distance of 1/2 inch or more.		X
	Tube portion of weatherstrip caught in slide fastener assembly stitching, in bartacks at base opening or in bartacks for attaching male snap fastener webbing retainers.		X
	Not assembled as specified.		X

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Examine	Defect	Classification Major	Minor
Notch reinforcement tape at base of front opening	Not stitched on all four sides of reinforcement.		X
	Set on outside of bag.		X
Foot straps	Single straps used instead of pairs.		X
	Free ends of straps not finished as specified.		X
	One or both bartacks reinforcing foot seam through foot straps omitted or misplaced, failing to serve intended purpose.		X
	NOTE: The bartacks are not visible from outside of foot seam. Their position can be determined by feeling foot seam.		

4.4.4 End item dimensional examination. The end item shall be examined for the defects listed below. The lot size shall be expressed in units of sleeping bags. The sample unit shall be one sleeping bag. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 4.0 for major defects and 15.0 for total (major and minor combined) defects.

Examine	Defect	Classification Major	Minor
Face opening	Head seam of outer and inner ply not finished seam on seam by more than 1/4 inch		X
Front opening	Top ends uneven by more than 5/8 inch (with slide fastener fully closed).		X
Flap	Foot end does not finish within the specified tolerance relative to the ends of the slide fastener tapes.		X
Snap fastener(s)	Any female snap fastener mislocated on flap in excess of tolerance but not more than 1/8 inch		X

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Examine	Defect	Classification Major	Minor
Snap fastener(s) (cont'd)	Any female snap fastener mislocated on flap in excess of tolerance by more than 1/8 inch causing inability to secure snap fastener without causing excessive puckering of flap.	X	
	Any male portion of snap fastener mislocated to the degree that female portion of snap fastener on flap cannot be engaged without causing excessive puckering. 1/	X	
	1/ For this defect all snaps on the flap shall be secured prior to performing the examination.		
Slide fastener	Outer edges of chains:		
	- protrude up to 1/16 inch beyond edge of guard strip or foundation strip		X
	- protrude more than 1/16 inch beyond edge of guard strip or foundation strip	X	
	NOTE: Defects to be scored only if condition exists for a distance of 8 inches or more or in several areas with an accumulated distance of 12 inches or more.		
	Lower edge of stop at bottom end of chain located:		
	- less than 1 inch but not less than 1/2 inch below notch at foot end of casing		X
	- less than 1/2 inch below notch at foot end of casing	X	
Foot straps	Finished length (for type I) less than 24-1/2 inches or more than 25-1/2 inches		X
	Finished length (for type II) less than 29-1/2 inches or more than 30-1/2 inches		X

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Examine	Defect	Classification Major	Minor
Foot straps (cont'd)	Located less than 4-1/4 inches or more than 4-3/4 inches from center of foot end of bag to edge of foot		
	Loops on inside of bag less than 3/4 inch or more than 1-1/4 inches in length.		X
Weatherstrip	Foot end extends below notch at foot end of bag:		
	- less than 2 inches but not less than 1 inch.		X
	- less than 1 inch	X	
Notch reinforcement tape at foot opening	Less than 2-3/4 inches or more than 3-1/4 inches in length.		X
	Not centered on seam at foot end by more than 1/4 inch		X
Channels	Any channel width exceeds tolerance:		
	- up to 1/4 inch		X
	- by more than 1/4 inch	X	
	NOTE: Defects to be scored only when condition exists for 8 inches or more, or in several areas with an accumulated distance of 12 inches or more. Applicable to each individual seam.		
Stitch margin or gage (not otherwise classified herein)	Exceeds specified tolerance.		X
	NOTE: Defect to be scored only when condition exists on major portion of seam. Applicable to each individual seam.		
Component and location dimensions not otherwise classified herein	Not within specified tolerance.		X

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4.4.5 End item weight examination. The end item shall be examined for the defects listed below. The lot size shall be expressed in units of sleeping bags. The sample unit shall be one sleeping bag. The inspection level shall be I and the AQL, expressed in terms of defects per hundred units, shall be 1.0 for major defects and 4.0 for total (major and minor combined) defects.

Examine	Defect	Classification Major	Minor
Completed sleeping bag for required weight of down 1/	Under the required total weight of down by 1 ounce or more		X
	Under the required total weight of down by 1/2 ounce up to but including 1 ounce		X

1/ The weight of the down shall be as specified in table I, operation 26. To determine weight of the down in the casing outer panels deduct the weight of the casing noted from total weight of filled bag. Utmost care shall be taken to assure that the apparatus used to perform the weight examination is properly calibrated and accurate (see 3.8).

4.4.6 Packaging inspection. An examination shall be made to determine that the preservation, packing and marking comply with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully packaged with the exception that it need not be closed. Examination of closure defects listed below shall be made on shipping containers fully packaged. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5 defects per 100 units.

Examine	Defect
Marking (exterior and unit pack)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Any component missing, damaged, or not as specified.
Workmanship	Inadequate application of components, such as incomplete closure of container flap, loose strapping, improper taping or inadequate stapling. Bulged or distorted container. Open or noncontinuous heat-sealed seams and closures of polyethylene bags. Incorrectly fabricated polyethylene bag.
Content	Number per container is more or less than required.

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4.4.7 Palletization examination. An examination shall be made to determine that the palletization complies with the section 5 requirements. Defects shallbe scorewd in accordance with the list below. The sample unit shall be one palletized unit load fully packaged. The lot size shall be the number of palletized unit loads in the end item inspection lot. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirements.
Palletization	Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.

4.5 Methods of inspection.

4.5.1 Component testing of labels (heat-sealed).

4.5.1.1 Preparation of samples for component testing. A label of the required size conforming to 3.3.14 shall be heat-sealed to a single ply of the cotton balloon cloth material specified in 3.3.1. The warp thread directions of the label material and cotton balloon cloth shall be aligned. The label shall be heat-sealed to the cotton balloon cloth material using the same type of equipment and under the same conditions of time, temperature and pressure as to be used in production of the sleeping bag. One end of the label shall be left unsealed a sufficient distance to provide tabs for subsequent testing. After heat-sealing the sample shall be cooled for a minimum of 10 seconds and then carefully removed from the heat-sealing apparatus without stressing the bonded area. The sample shall then be allowed to remain at room temperature for 1 hour minimum. The sample prepared as specified above shall be prepared for testing by cutting the sample into strips 1 $\pm$ 1/32 inch wide along the long dimension of the label.

4.5.1.2 Adhesion test. Five 1 $\pm$ 1/32 inch wide specimens, prepared as specified in 4.5.1.1, shall be tested for adhesion in accordance with Method 5960 of FED-STD-191.

4.5.1.3 Hot-soap resistance test. Five 1 $\pm$ 1/32 inch wide specimens, prepared as specified in 4.5.1.1, shall be tested for hot-soap resistance. The five specimens shall be submerged in 500 mL of a 1/4 percent by weight solution of high titer laundry soap conforming to type I, class 1 or 2 of P-S-1792, maintained at 160°F + 5°F for 1 hour minimum. The specimens shall then be rinsed in a beaker kept overflowing with hot tap water (130° to 140°F) for approximately 2 minutes, and then shall be given a final rinse for approximately 2 minutes in water maintained at a temperature of 70°F $\pm$ 5°F. The specimens shall then be tested while wet, within 1 hour, for adhesion in accordance with Method 5960 of FED-STD-191.

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4.5.2 Intermediate testing of labels heat-sealed to sleeping bag during production.

4.5.2.1 Preparation of specimens for intermediate testing. A 1 ± 1/32 inch wide specimen shall be cut along the long dimension from labels heatsealed to sleeping bag inner back panels or unfilled casings (see 4.4.3). More than one specimen may be cut from each back panel or unfilled casing. The ends of the specimens shall be separated to provide tabs of sufficient length to perform the adhesion test specified in 4.5.2.2. The separation may be effected by hand immediately after heating, the area which is to be separated, with a hot iron. Only the area to be separated shall be heated.

4.5.2.2 Intermediate adhesion test. Five 1 ± 1/32 inch wide specimens, prepared as specified in 4.5.2.1, shall be tested for adhesion in accordance with Method 5960 of FED-STD-191.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Folding. Each sleeping bag, with the slide fastener closed, shall first be laid out flat and then folded in half with the fold running the length of the bag, The hood shall be positioned within the sleeping bag at the head end. Each sleeping bag shall then be folded twice in half from head to foot.

5.1.1.2 Unit pack. Each folded sleeping bag shall be inserted in a snugfitting flat clear polyethylene film bag of 0.003 inch thickness (± 20 percent tolerance). The bag may be fabricated from polyethylene film tubing or sheeting. The polyethylene bag shall be formed with heatsealed seams that are straight, continuous and parallel to each other and the formed edges of the bag. The bag closure shall be effected by heatsealing with the heat-seal made as close as possible to the open end. Prior to or during the final heat-sealing closure operation, excess air within the bag shall be expelled.

* 5.1.2 Commercial. The sleeping bags shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B or Commercial as specified (see 6.2).

* 5.2.1 Level A packing. Four sleeping bags, preserved as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC-L, grade V2s of PPP-B-636. The inside of each container shall be fitted with a boxed liner conforming to type CF, class weather-resistant, variety DW, grade V15c of PPP-B-636. Level A unit packs shall be packed flat, four in depth within a shipping container, as an alternate, level A unit packs shall be packed on their sides, three in length and one in width within a shipping container. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V and reinforced as specified in the appendix of PPP-B-636, except the inspection shall be in accordance with 4.5.

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5.2.2 Level B packing. Four sleeping bags, preserved as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC-L, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. The inside of each container shall be fitted with a box liner conforming to type CF, class domestic, variety DW, grade 275 of PPP-B-636. Level A unit packs shall be packed flat, four in depth, within a shipping container. Each container shall be closed in accordance with method II as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.6.

5.2.2.1 Weather resistant fiberboard containers. When specified (see 6.2), the shipping container shall be a grade V3c, V3s or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of the container specification, except that the inspection shall be in accordance with 4.4.6.

5.2.3 Commercial packing. Sleeping bags, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Palletization. When specified (see 6.2), sleeping bags, packed as specified in 5.2, shall be palletized in accordance with load type I of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with the bonding means K and L. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course. In the container is of a size which does not conform to any of the patterns specified in MIL-STD-147, the pallet pattern used shall first by approved by the contracting officer.

5.4 Marking. In addition to any special marking required by the contract or purchase order, unit packs and shipping containers shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

5.4.1 Polyethylene bagged unit packs. Polyethylene bagged unit packs shall have the required identification information legibly printed or stamped in black directly on the bag across the center face or on a white paper label inserted within the bag so as to permit ready identification. The required information shall be as follows:

NSN
Nomenclature
Contract No.

6. NOTES

6.1 Intended use. The sleeping bag is for use in locations where mean monthly temperatures are below +10°F.

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6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. When a first article sample is required (see 3.2, 4.3 and 6.4).
c. Selection of applicable levels of preservation and packing (see 5.1 and 5.2).
d. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).

6.3 Samples. For access to samples, address the contracting activity issuing the invitation for bids.

6.4 First article. When a first article sample is required, it shall be inspected and approved under the appropriate provisions of DAR 7-104.55. The first article should be a preproduction sample consisting of one completed sleeping bag and one unfilled casing. The contracting officer should include specific instructions in all acquisition documents for inspection and approval of the first article.

6.5 Direct marking ink. Suggested ink for applying direct markings is the Markem 8272 black ink from the Markem Co., Keene, NH.

6.6 Non-woven cloth sources of supply. Source of supply for the non-woven cloth specified in 3.3.2 are the Monsanto Company, New Enterprise Division, St, Louis, MO 63166.

6.7 Recycled materials. It is encouraged that recycled material be used when practical as long as it meets the requirements of the document (see 3.3).

6.8 Changes from previous issue. The margins of this document are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liaby whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Navy - NU
Air Force - 99

Preparing activity:

Army - GL

Project No. 8465-0877

Review activities:

Army - MD
Navy - MC
Air Force - 11, 82
DLA - CT

User activity:

Air Force - 45

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-43880C	Sleeping Bag, Extreme Cold Weather (Down and Polyester Batting)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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